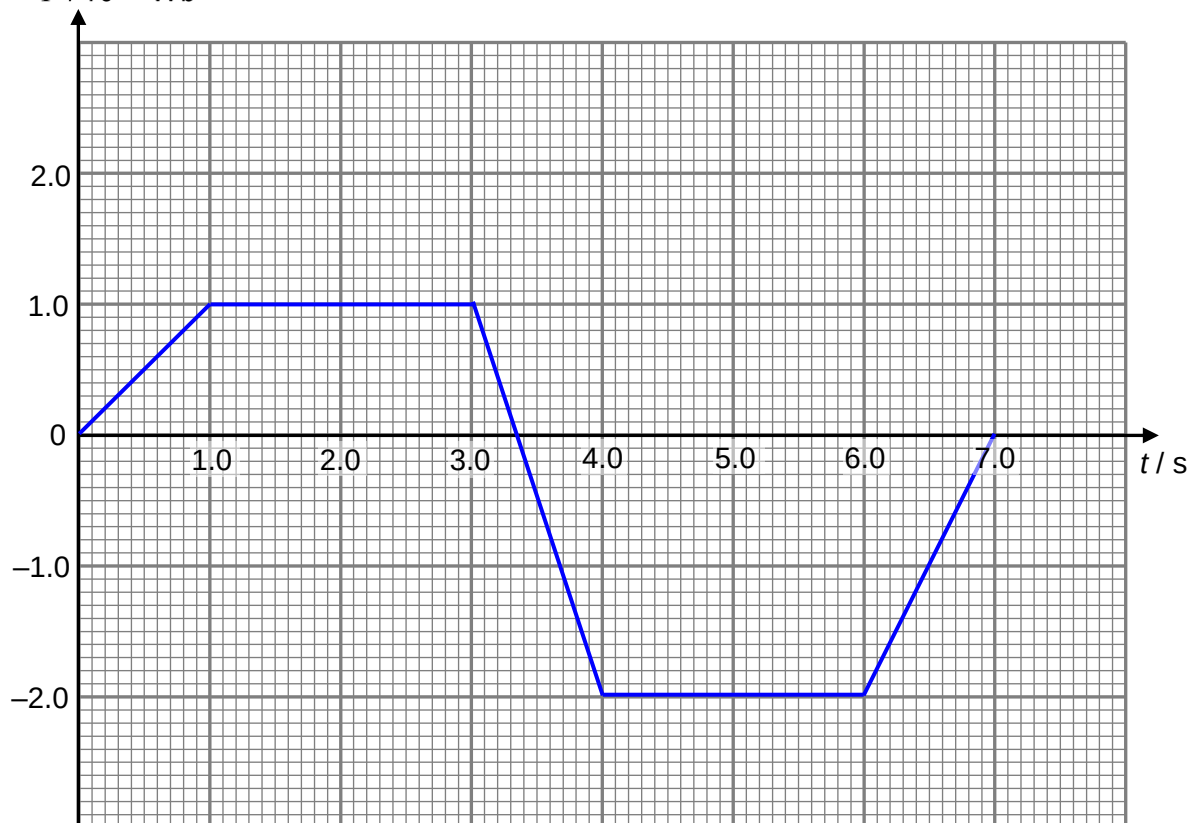


5

(a) $\Phi / 10^{-5} \text{ Wb}$



- (b) By Faraday's Law, an e.m.f. is induced in the coil when there are changes in the magnetic flux linkage of the coil as it moves through the fields.

By Lenz's Law, the induced e.m.f. causes a current to flow around the coil such that it induces a magnetic force that acts on the coil in the direction opposite to the direction of the coil's velocity. This magnetic force causes the coil to decelerate.

Hence, in order to maintain constant velocity, an external force of the same magnitude as the magnetic force needs to be applied in the direction of the coil's displacement, which results in work done on the coil.

- (c) The maximum e.m.f. is induced when there is a maximum rate of change of magnetic flux i.e. when it is transitioning from P to Q, where the gradient of the flux-time graph is the steepest.

By Faraday's Law,

$$\begin{aligned}
 |E_{\max}| &= \left| -\frac{d\Phi}{dt} \right| \\
 &= \left| -\frac{(-2.0 \times 10^{-5}) - (1.0 \times 10^{-5})}{4.0 - 3.0} \right| \\
 &= 3.0 \times 10^{-5} \text{ V}
 \end{aligned}$$

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6 **(a)** The photoelectric effect is only observed when the frequency of the electromagnetic